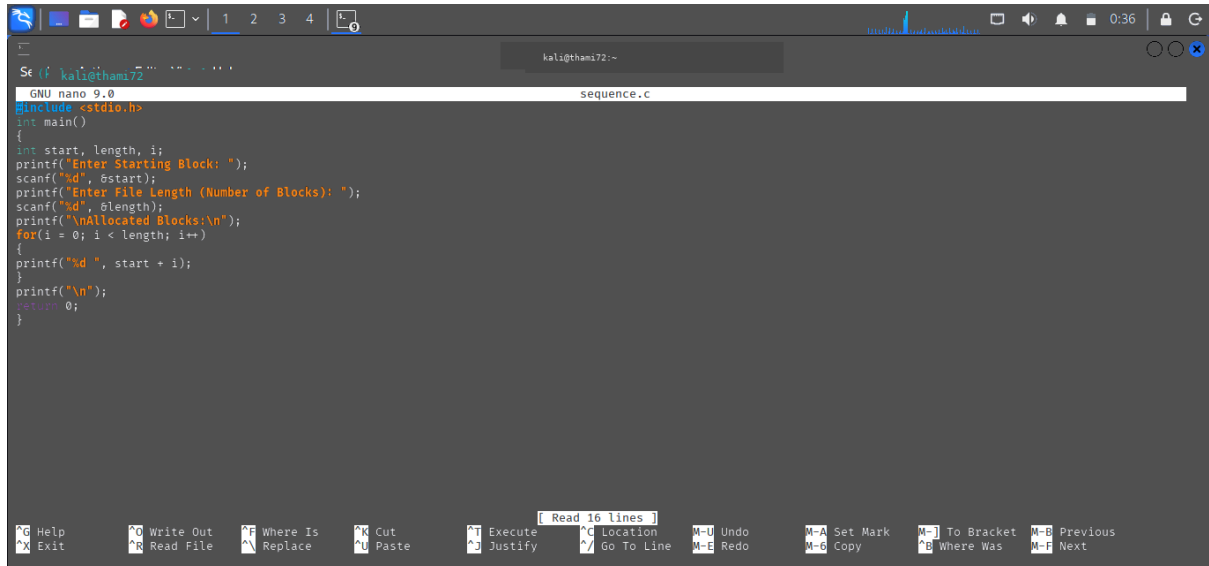


C PROGRAMMING :SEQUENTIAL

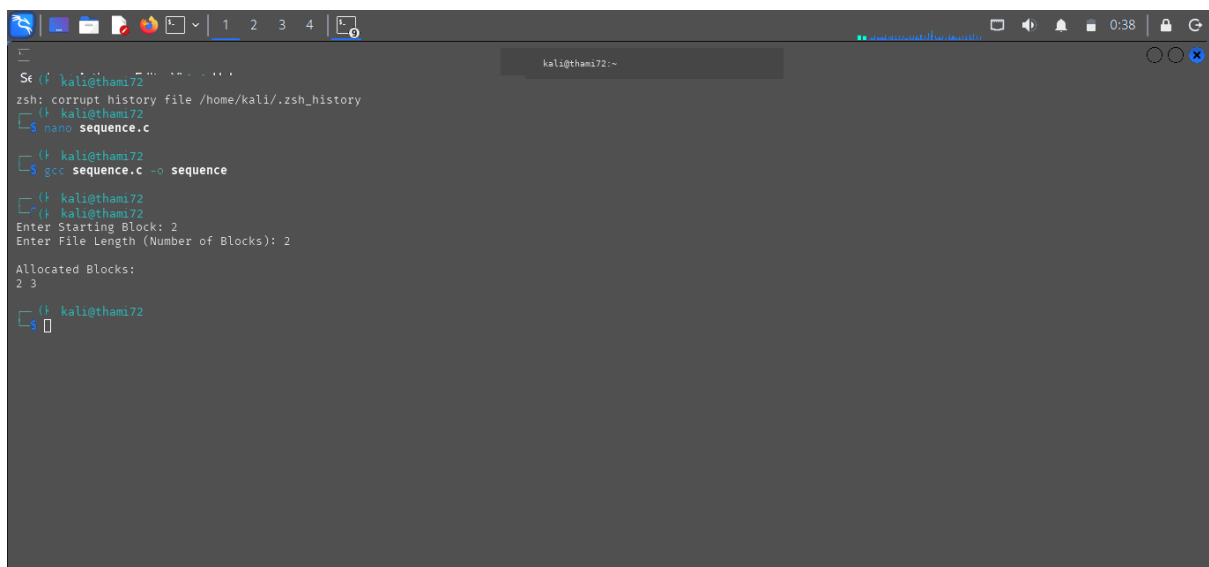


```
Se (f kali@thami72 ~)
GNU nano 9.0 sequence.c
#include <stdio.h>
int main()
{
    int start, length, i;
    printf("Enter Starting Block: ");
    scanf("%d", &start);
    printf("Enter File Length (Number of Blocks): ");
    scanf("%d", &length);
    printf("\nAllocated Blocks:\n");
    for(i = 0; i < length; i++)
    {
        printf("%d ", start + i);
    }
    printf("\n");
    return 0;
}
```

Read 16 lines

Help Exit Write Out Read File Where Is Replace Cut Paste Execute Justify Location Go To Line Undo Redo Set Mark Copy To Bracket Where Was Previous Next

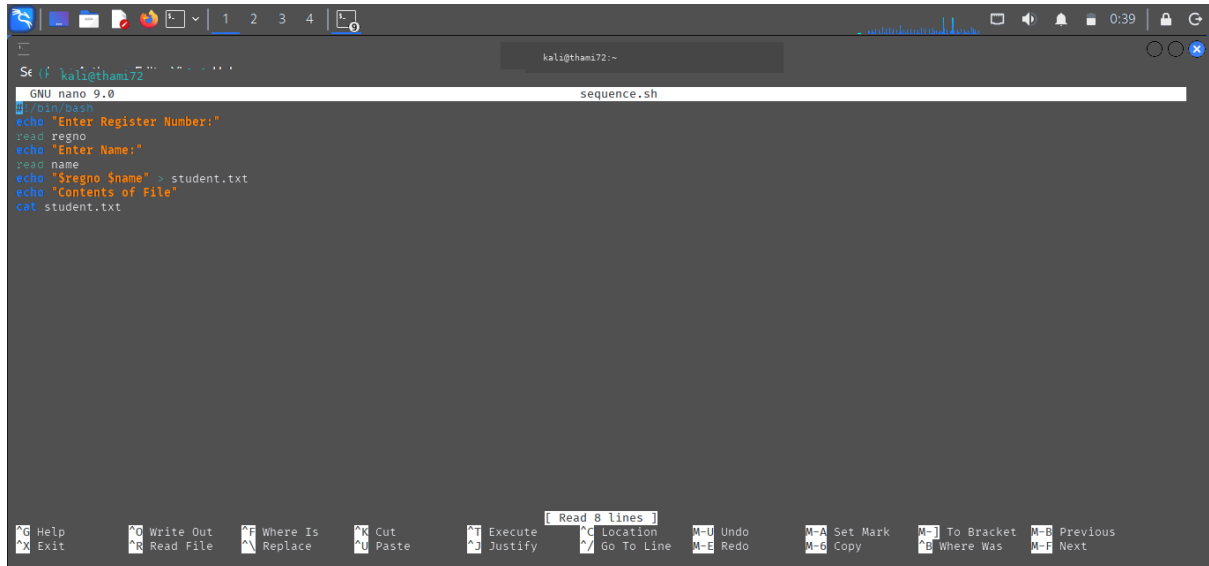
OUTPUT :



```
Se (f kali@thami72 ~)
zsh: corrupt history file /home/kali/.zsh_history
kali@thami72 ~$ nano sequence.c
kali@thami72 ~$ gcc sequence.c -o sequence
kali@thami72 ~$ ./sequence
Enter Starting Block: 2
Enter File Length (Number of Blocks): 2

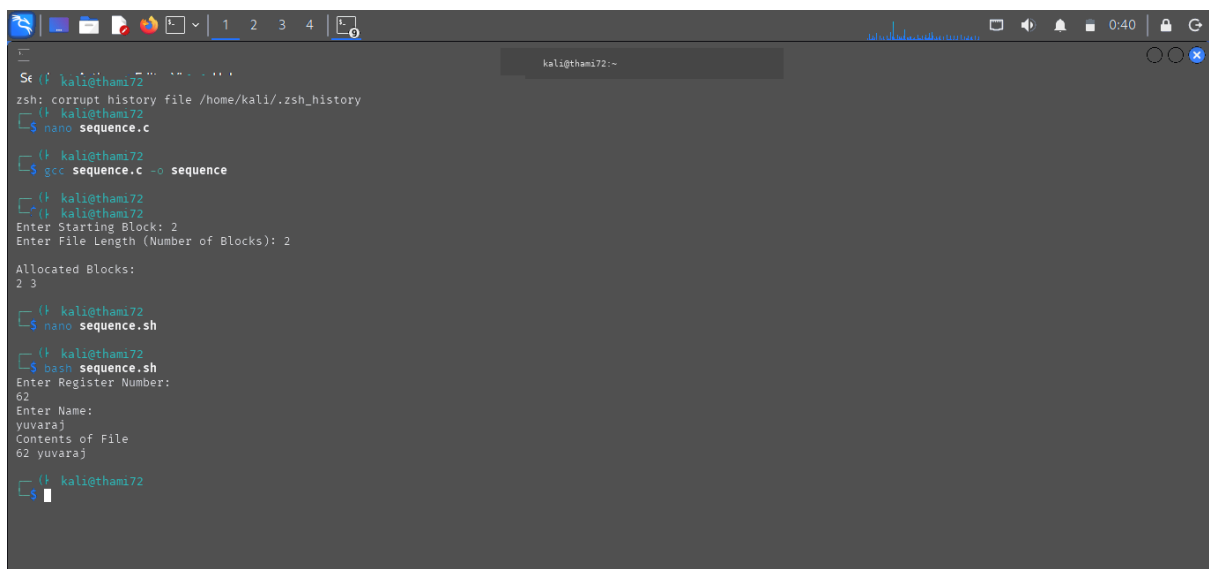
Allocated Blocks:
2 3
kali@thami72 ~$
```

SHELL PROGRAMMING :SEQUENTIAL



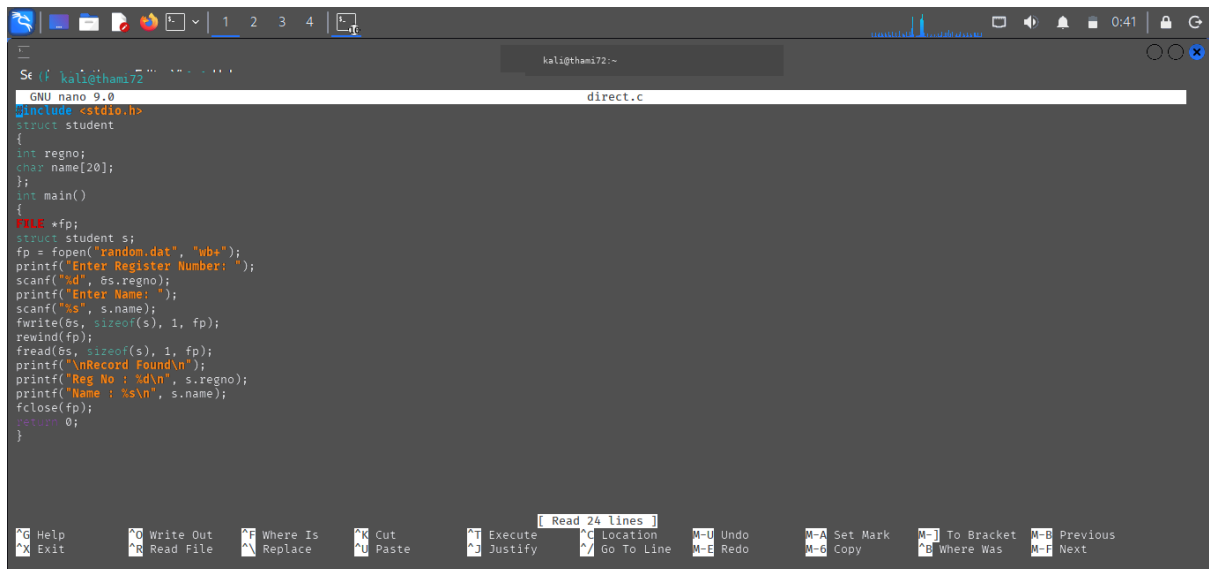
```
Se (f kali@thami72 ~)
GNU nano 9.0 sequence.sh
#!/bin/bash
echo "Enter Register Number:"
read regno
echo "Enter Name:"
read name
echo "$regno $name" > student.txt
echo "Contents of File"
cat student.txt
```

OUTPUT :



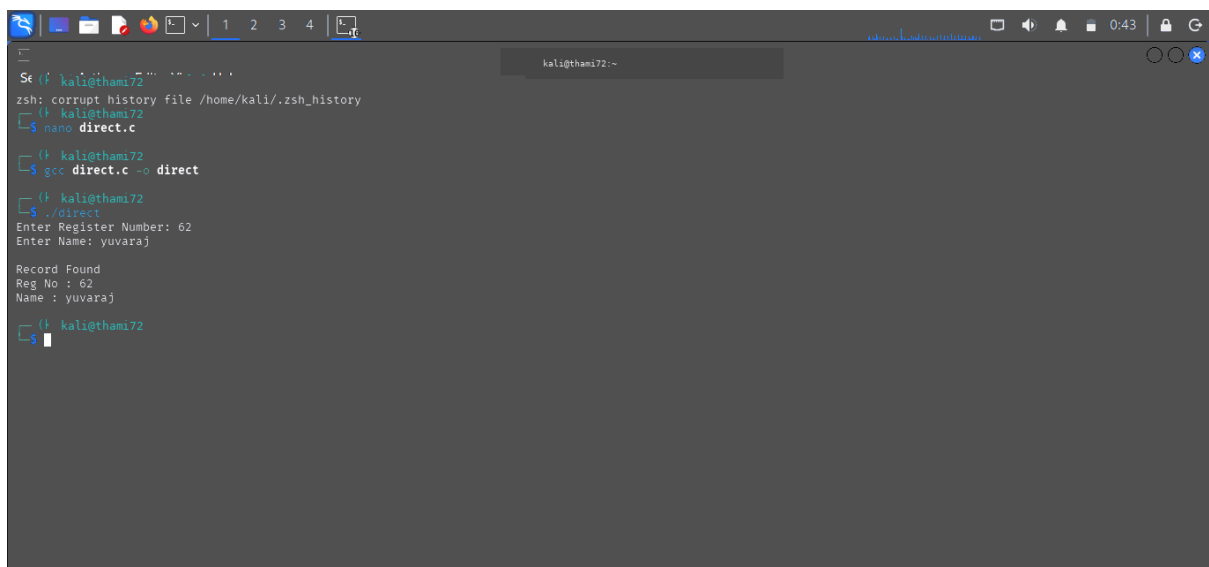
```
Se (f kali@thami72 ~)
zsh: corrupt history file /home/kali/.zsh_history
kali@thami72 ~$ nano sequence.c
kali@thami72 ~$ gcc sequence.c -o sequence
kali@thami72 ~$ ./sequence
Enter Starting Block: 2
Enter File Length (Number of Blocks): 2
Allocated Blocks:
2 3
kali@thami72 ~$ nano sequence.sh
kali@thami72 ~$ bash sequence.sh
Enter Register Number:
62
Enter Name:
yuvaraj
Contents of File
62 yuvaraj
kali@thami72 ~$
```

C PROGRAMMING :DIRECT



```
GNU nano 9.0 direct.c
#include <stdio.h>
struct student
{
    int regno;
    char name[20];
};
int main()
{
    FILE *fp;
    struct student s;
    fp = fopen("random.dat", "wb+");
    printf("Enter Register Number: ");
    scanf("%d", &s.regno);
    printf("Enter Name: ");
    scanf("%s", s.name);
    fwrite(&s, sizeof(s), 1, fp);
    rewind(fp);
    fread(&s, sizeof(s), 1, fp);
    printf("\nRecord Found\n");
    printf("Reg No : %d\n", s.regno);
    printf("Name : %s\n", s.name);
    fclose(fp);
    return 0;
}
```

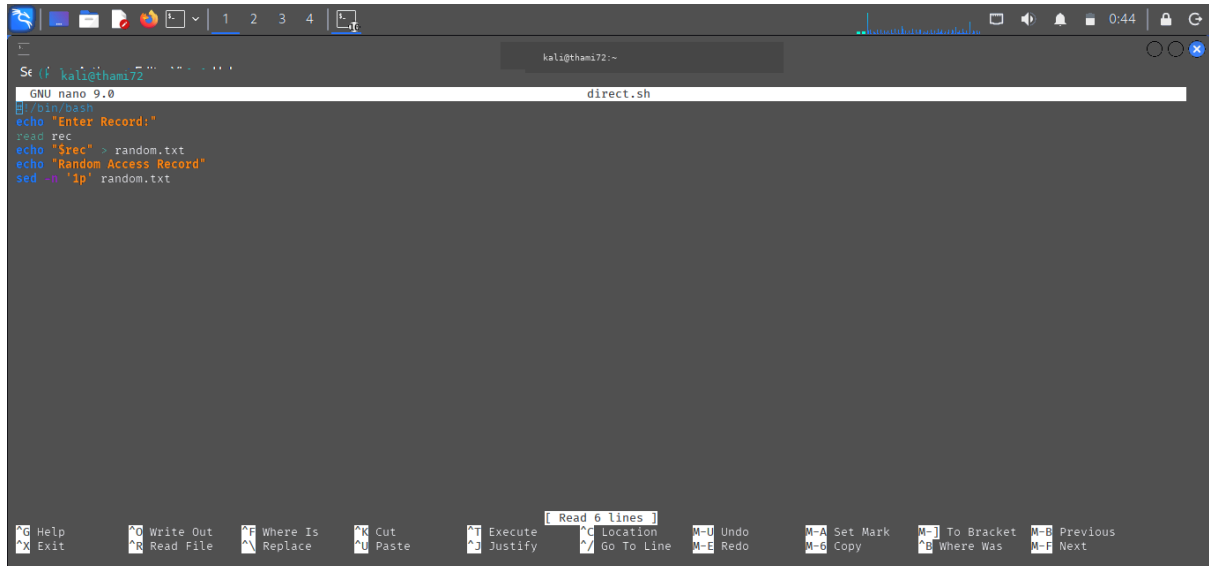
OUTPUT :



```
zsh: corrupt history file /home/kali/.zsh_history
kali@thami72 ~
$ nano direct.c
kali@thami72 ~
$ gcc direct.c -o direct
kali@thami72 ~
$ ./direct
Enter Register Number: 62
Enter Name: yuvaraj

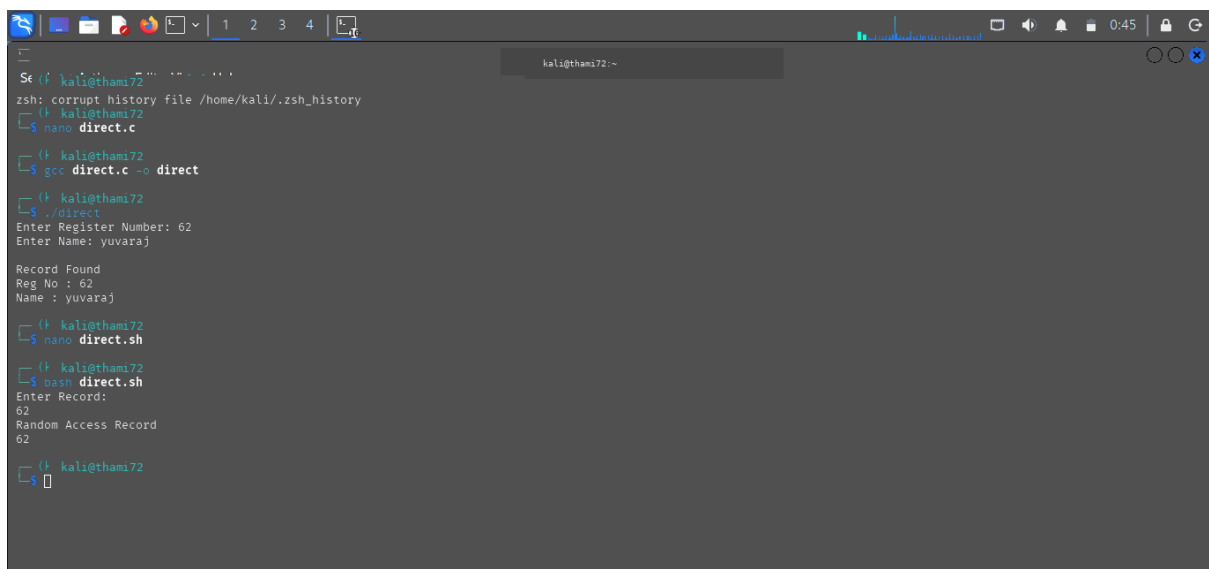
Record Found
Reg No : 62
Name : yuvaraj
kali@thami72 ~
$
```

SHELL PROGRAMMING :DIRECT



```
GNU nano 9.0 direct.sh
#!/bin/bash
echo "Enter Record:"
read rec
echo "$rec" > random.txt
echo "Random Access Record"
sed -n '1p' random.txt
```

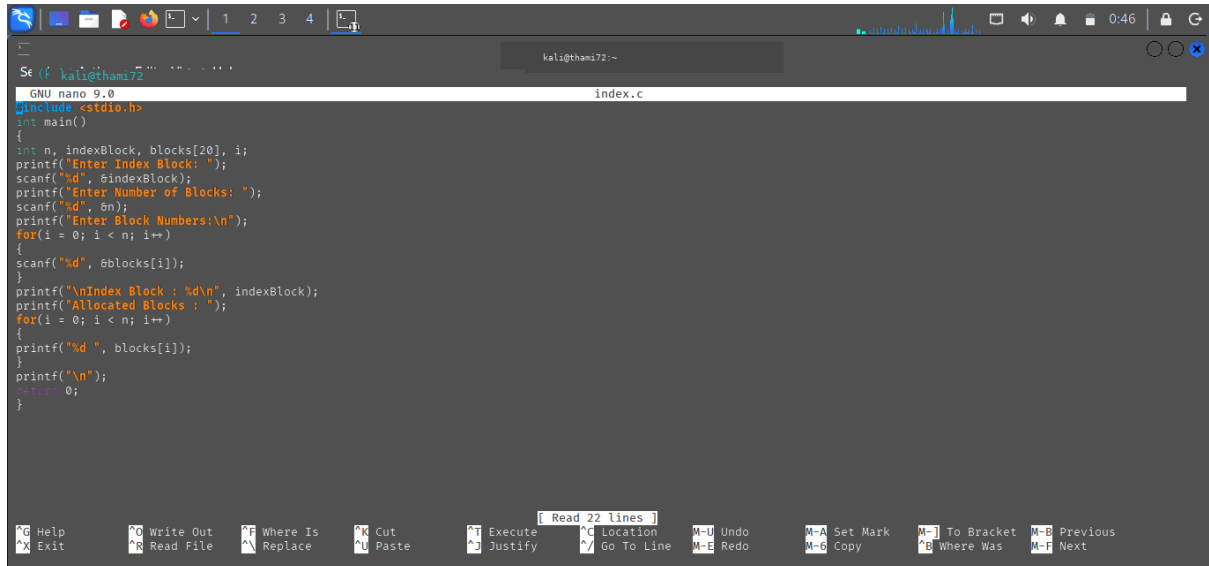
OUTPUT :



```
zsh: corrupt history file /home/kali/.zsh_history
kali@thami72 ~
$ nano direct.c
kali@thami72 ~
$ gcc direct.c -o direct
kali@thami72 ~
$ ./direct
Enter Register Number: 62
Enter Name: yuvaraj

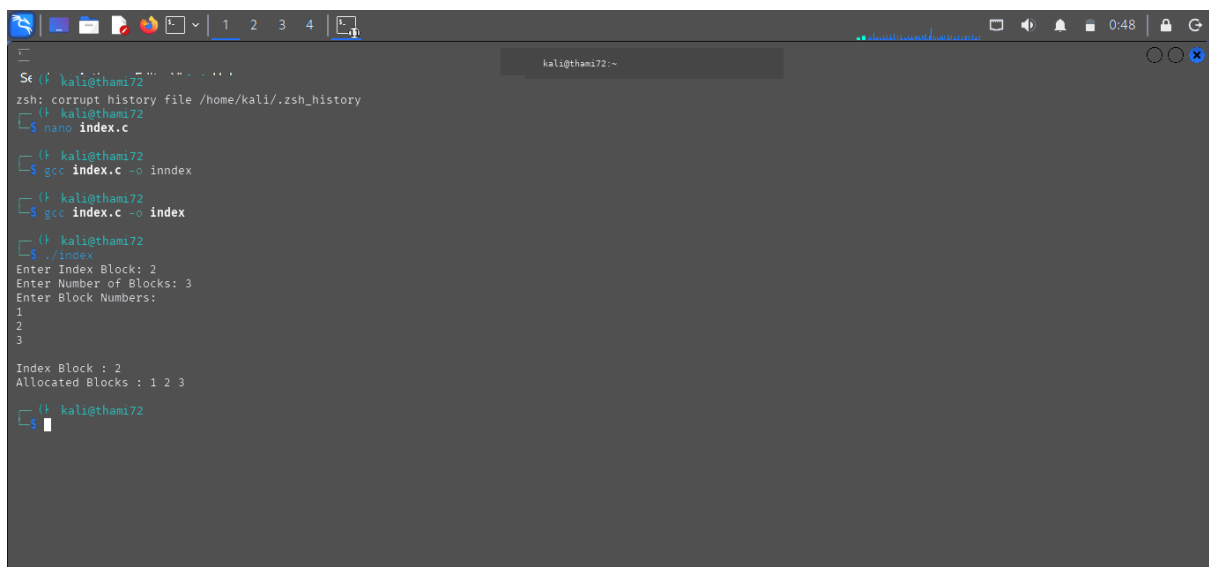
Record Found
Reg No : 62
Name : yuvaraj
kali@thami72 ~
$ nano direct.sh
kali@thami72 ~
$ bash direct.sh
Enter Record:
62
Random Access Record
62
kali@thami72 ~
$
```

C PROGRAMMING :INDEX



```
GNU nano 9.0 index.c
#include <stdio.h>
int main()
{
    int n, indexBlock, blocks[20], i;
    printf("Enter Index Block: ");
    scanf("%d", &indexBlock);
    printf("Enter Number of Blocks: ");
    scanf("%d", &n);
    printf("Enter Block Numbers:\n");
    for(i = 0; i < n; i++)
    {
        scanf("%d", &blocks[i]);
    }
    printf("\nIndex Block : %d\n", indexBlock);
    printf("Allocated Blocks : ");
    for(i = 0; i < n; i++)
    {
        printf("%d ", blocks[i]);
    }
    printf("\n");
    return 0;
}
```

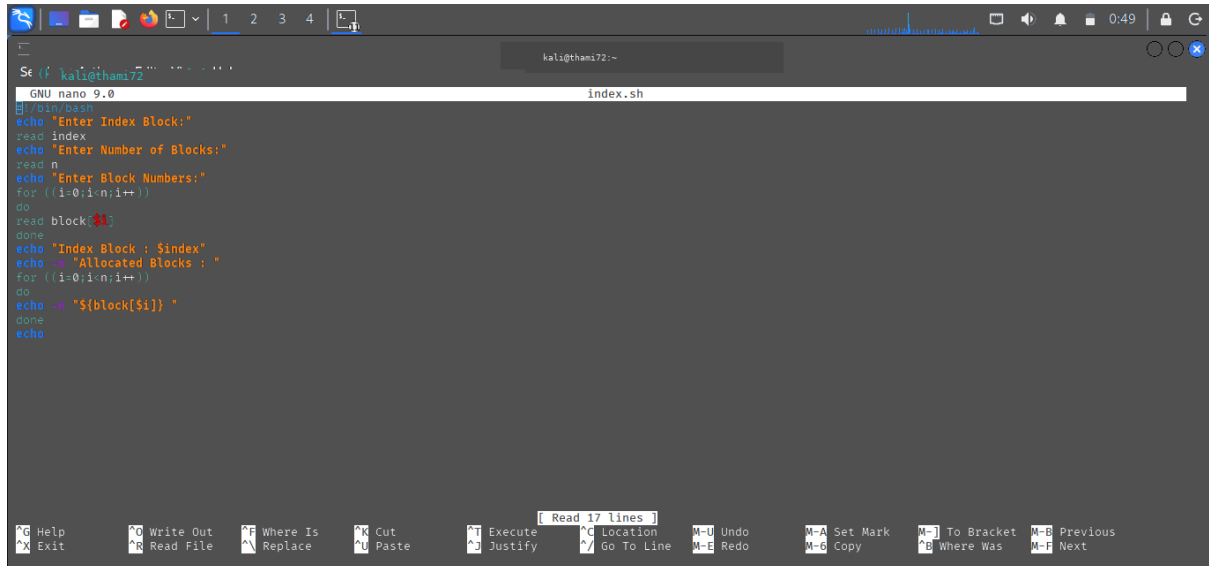
OUTPUT :



```
Se (f kali@thami72 ~)
zsh: corrupt history file /home/kali/.zsh_history
kali@thami72 ~
$ nano index.c
kali@thami72 ~
$ gcc index.c -o inindex
kali@thami72 ~
$ gcc index.c -o index
kali@thami72 ~
$ ./index
Enter Index Block: 2
Enter Number of Blocks: 3
Enter Block Numbers:
1
2
3

Index Block : 2
Allocated Blocks : 1 2 3
kali@thami72 ~
$
```

SHELL PROGRAMMING : INDEX

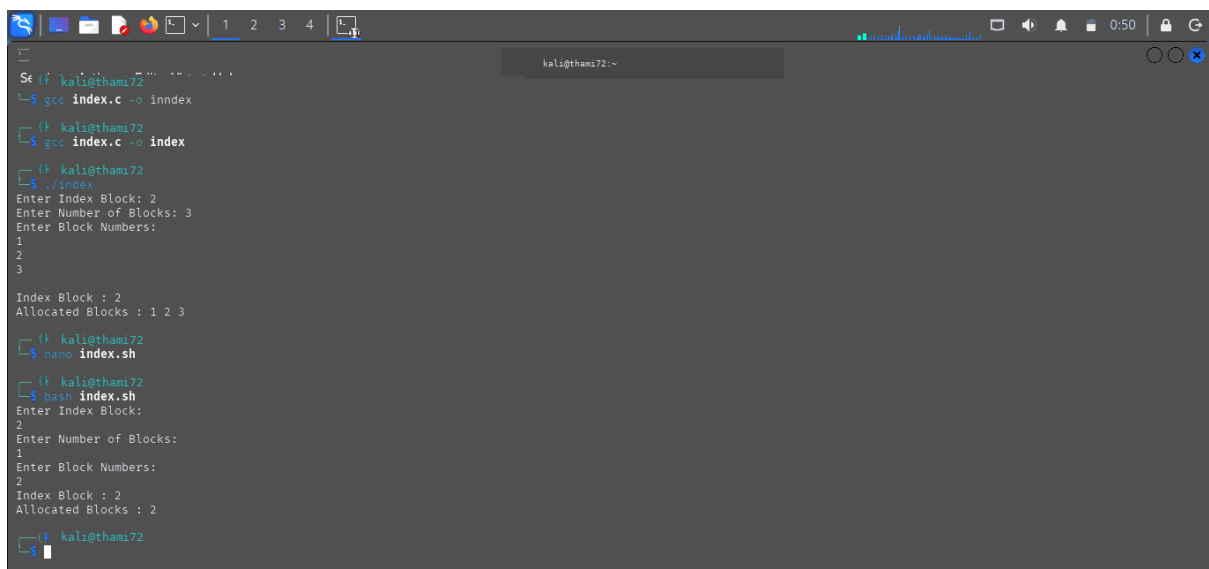


A terminal window on a Kali Linux system. The nano text editor is open, editing a file named `index.sh`. The script contains the following code:

```
#!/bin/bash
echo "Enter Index Block:"
read index
echo "Enter Number of Blocks:"
read n
echo "Enter Block Numbers:"
for ((i=0;i<n;i++))
do
read block${i}
done
echo "Index Block : $index"
echo -n "Allocated Blocks : "
for ((i=0;i<n;i++))
do
echo -n "${block[$i]} "
done
echo
```

The terminal window shows the command prompt `kali@thami72:~` and the file name `index.sh` in the title bar. The nano editor's status bar at the bottom indicates "Read 17 lines".

OUTPUT :



A terminal window showing the execution of the program. The user runs `gcc index.c -o inindex` and `gcc index.c -o index`. Then, they run `./index`, which prompts for input. The user enters `2` for the Index Block and `3` for the Number of Blocks. The program then prompts for Block Numbers, and the user enters `1`, `2`, and `3`. The output shows the Index Block and Allocated Blocks.

```
kali@thami72:~$ gcc index.c -o inindex
kali@thami72:~$ gcc index.c -o index
kali@thami72:~$ ./index
Enter Index Block: 2
Enter Number of Blocks: 3
Enter Block Numbers:
1
2
3

Index Block : 2
Allocated Blocks : 1 2 3

kali@thami72:~$ nano index.sh
kali@thami72:~$ dbash index.sh
Enter Index Block:
2
Enter Number of Blocks:
1
Enter Block Numbers:
2
Index Block : 2
Allocated Blocks : 2

kali@thami72:~$
```